Embracing Serverless: reengineering a real-estate digital marketplace

The journey of transitioning an on-premises monolithic real estate application to a fully serverless, event-driven microservices stack on AWS.

BATBern 10.11.2023

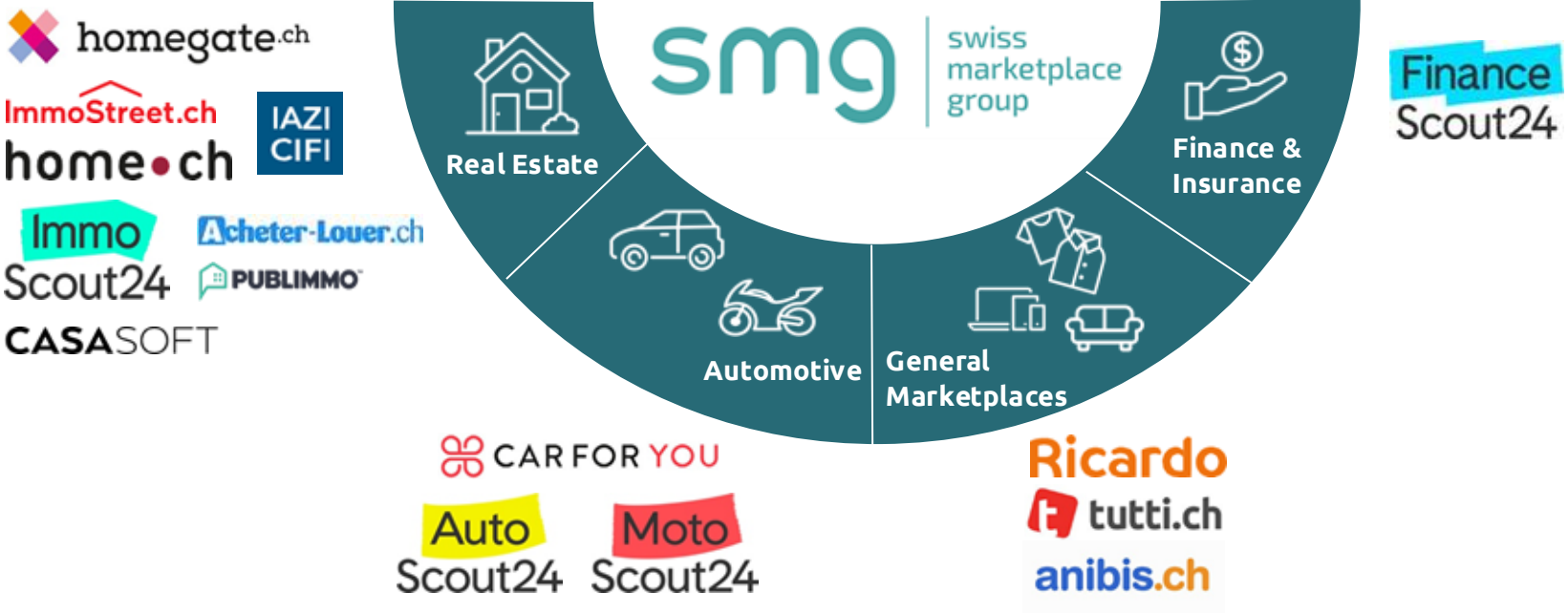
swiss marketplace group

About me

Guido Trüeb

Principal Cloud Engineer @ SMG real-estate

Building and breaking software since 2005, I've embraced serverless architectures at homegate.ch since 2019. A firm believer in simplicity and compassionate teamwork, I relish optimizing software and collaboration - particularly fondly of removing code.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ImmoStreet.ch
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Immo Scout24
Acheter-Louer.ch
PUBLIMMO
CASASOFT
smg
swiss marketplace group
Real Estate
Finance & Insurance
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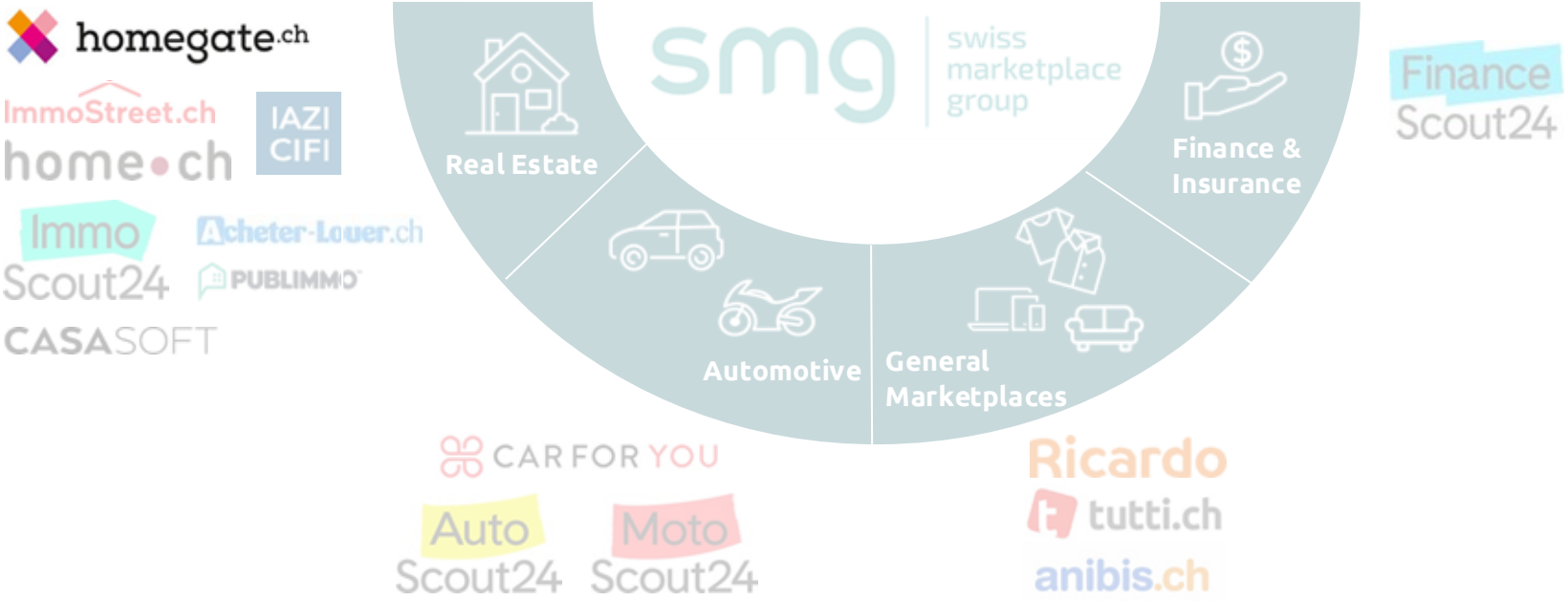
About us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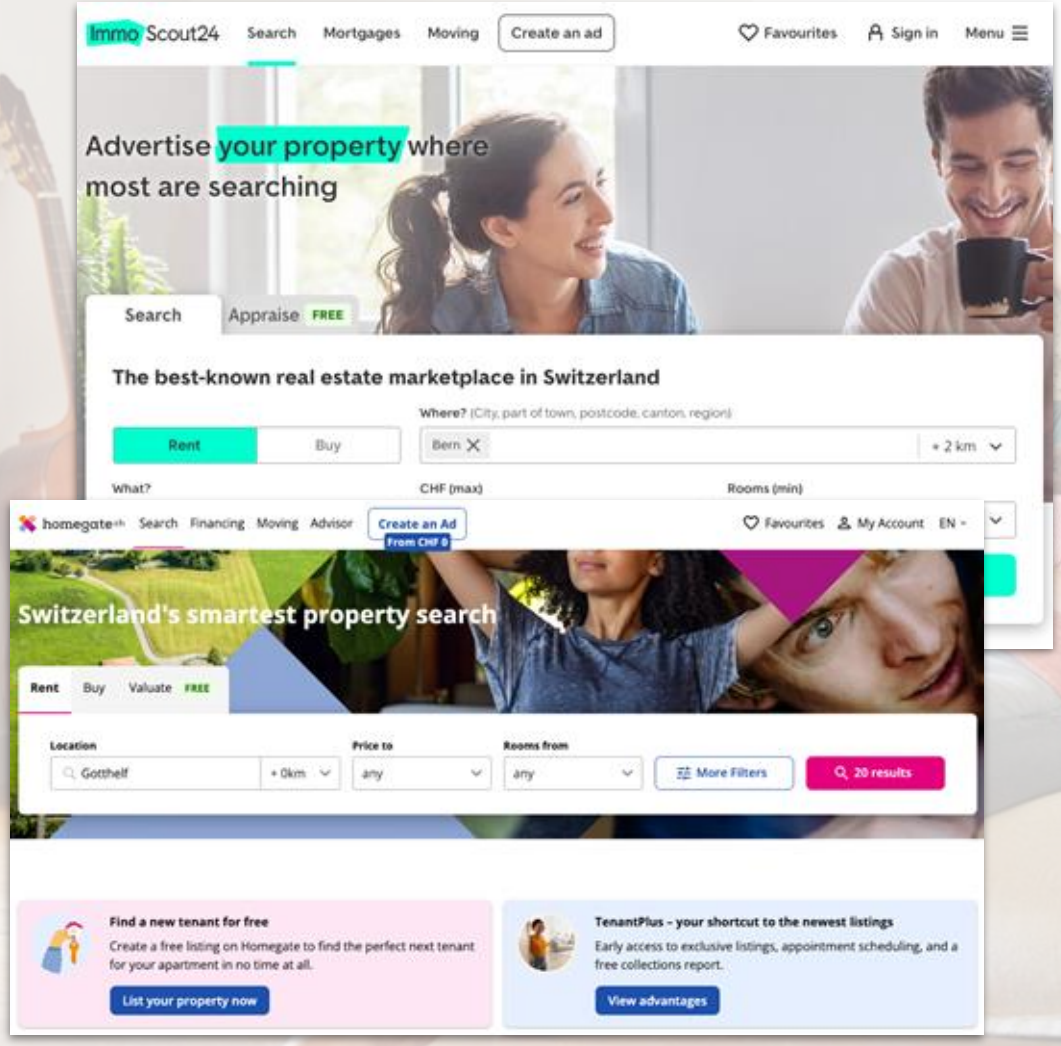
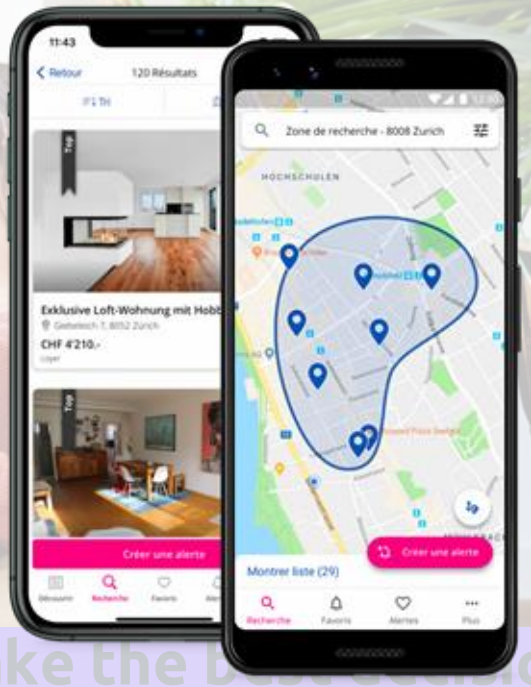
We empower
everyone to
make the best decisions
for Swiss real estate,
digital and simple





make the best decisions

for Swiss real estate,

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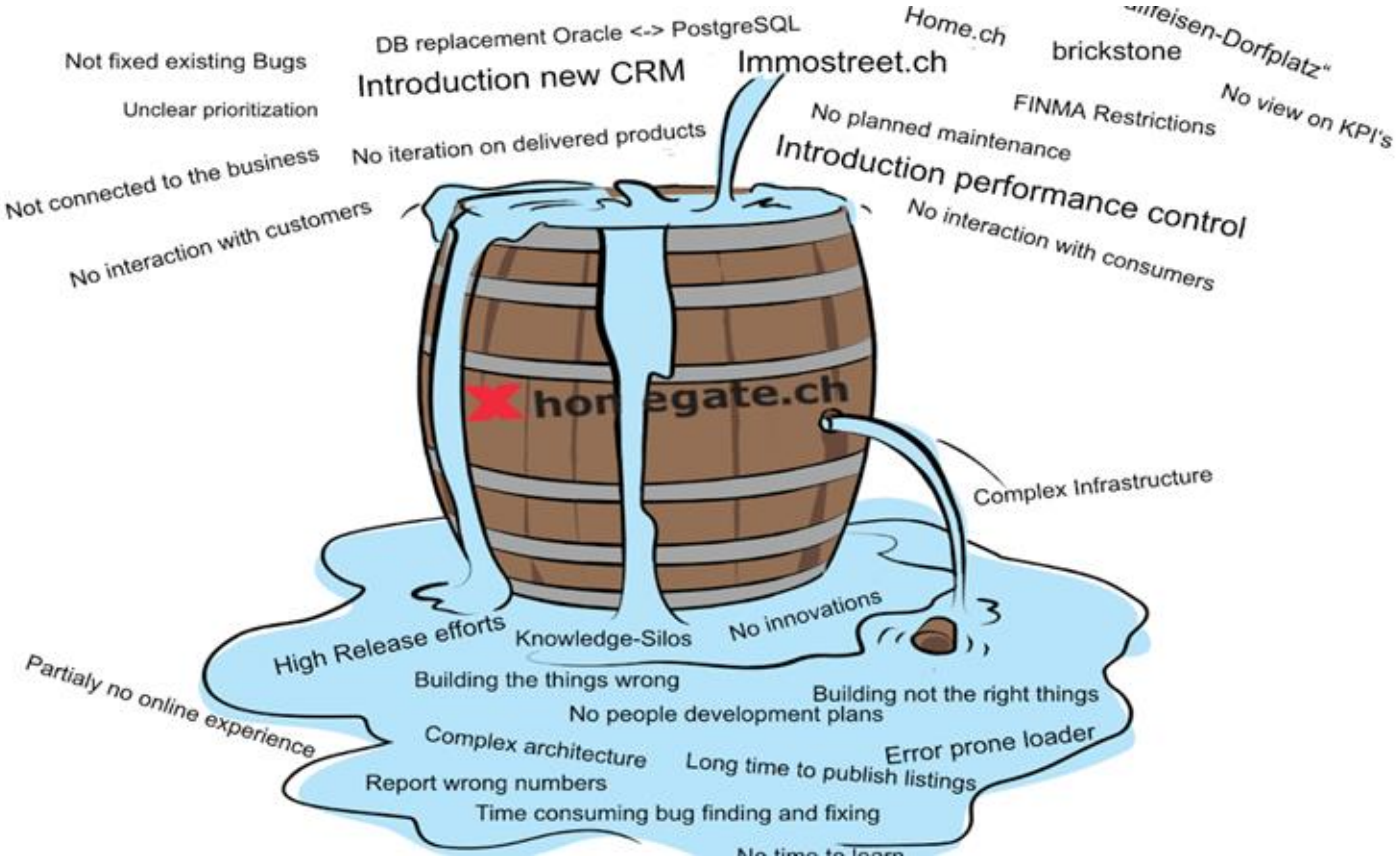
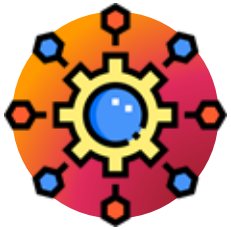

Not fixed existing Bugs
DB replacement Oracle <-> PostgreSQL
Home.ch
brickstone
Raiffeisen-Dorfplatz"
Immostreet.ch
Introduction new CRM
Unclear prioritization
No view on KPI's
FINMA Restrictions
No planned maintenance
No iteration on delivered products
Not connected to the business
Introduction performance control
No interaction with customers
No interaction with consumers
homegate.ch
Complex Infrastructure
High Release efforts
Knowledge-Silos
No innovations
Partialy no online experience
Building the things wrong
Building not the right things
No people development plans
Complex architecture
Error prone loader
Long time to publish listings
Report wrong numbers
Time consuming bug finding and fixing

Pressure from competition...

...let's **bring value faster**!

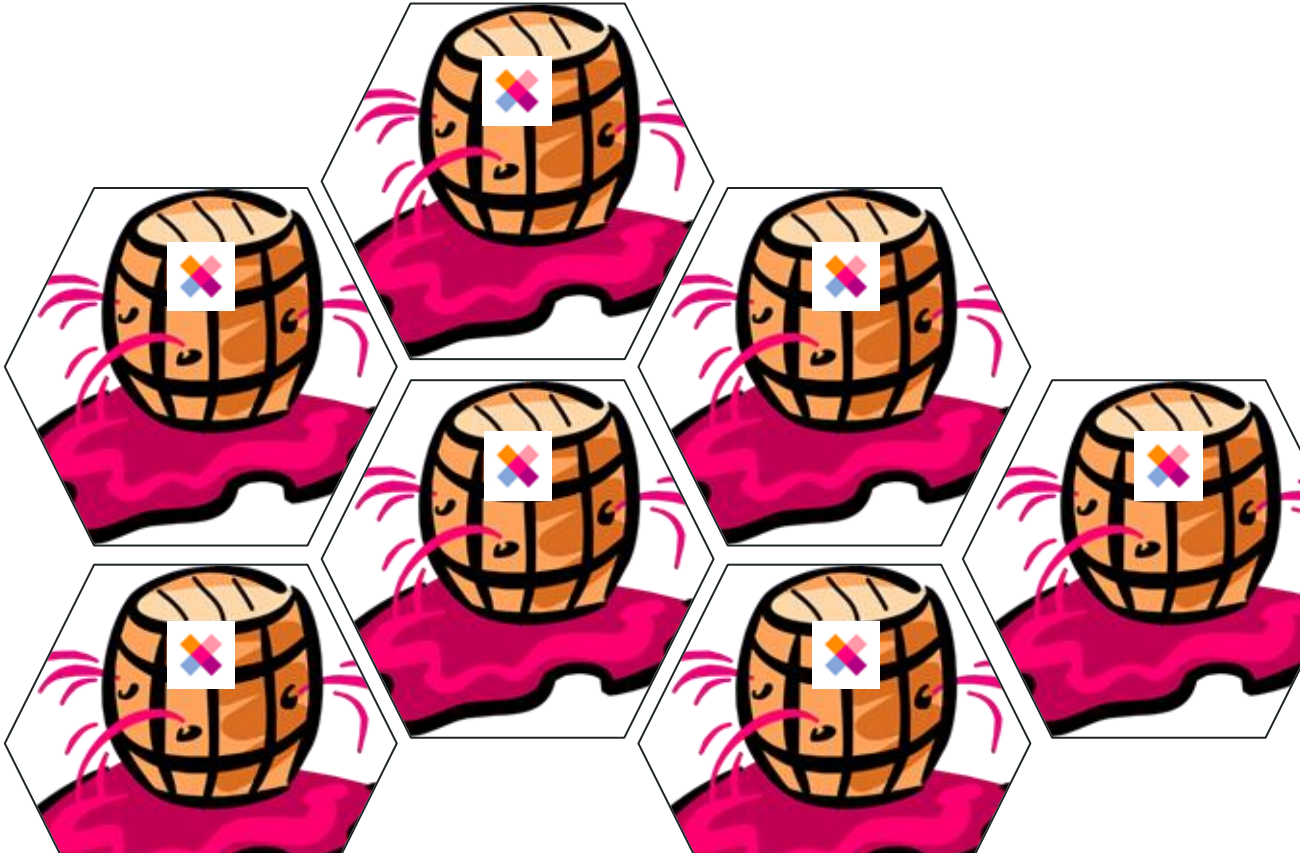
Attract talents
(culture + tech stack)



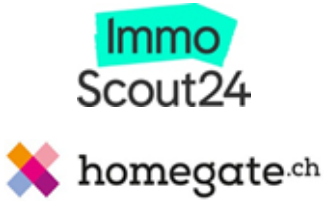
Scale the team
(event-driven microservices)

Focus on core business
(serverless)

4 years later...



5 years later...



We are a team of over **90 people**, divided into **13 teams**, managing around **50 services and applications**. Out of these, **15 are critical** for maintaining the core user journeys.

Immo Scout24

homegate.ch

~2 million page views per day

~1 million active search subscriptions generating **~2 million notifications daily**

Seasonal peaks on March and October, besides the **daily and weekly fluctuations**.

* numbers from homegate.ch only

Disclaimers

Serverless in our context

"Build and run applications without thinking about servers."

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- running code
- managing data
- integrating applications

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"Build and run applications without thinking about servers."

- running code
- managing data
- integrating applications
- automatic scaling
- built-in high availability
- pay-for-use billing model

Highly-distributed solutions expose complexity and limitations

For example

DynamoDB query capabilities

- Data distributed by partition keys
- Indexes characteristics/limitations

SNS/SQS at-least-once out-of-order delivery

- Listeners must handle events idempotently

S3 eventual consistency and storage classes

Serverless scales, but…

Downstream systems do not scale endlessly.

- AWS services have requests limits or other quotas
- 3rd party services have requests limits

Be aware of service characteristics and its limitations.

Vast portfolio of services

A lot of building blocks **ready to be assembled to solve your problems**

A lot of building blocks **to choose from in your design (analysis paralysis)**

Every service has **useful and capable features**

Every service has **limitations**

Features and services **introduced weekly**

Features and services **to be learned weekly**

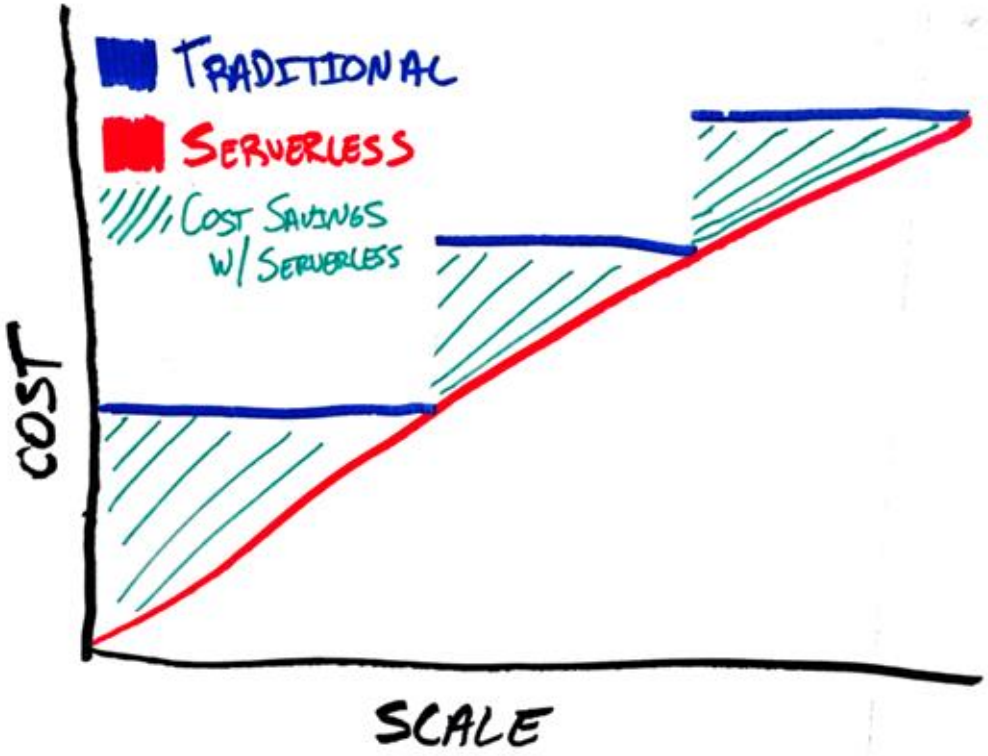
Freshly released features **solving real problems**

Freshly released features **sometimes unpolished**

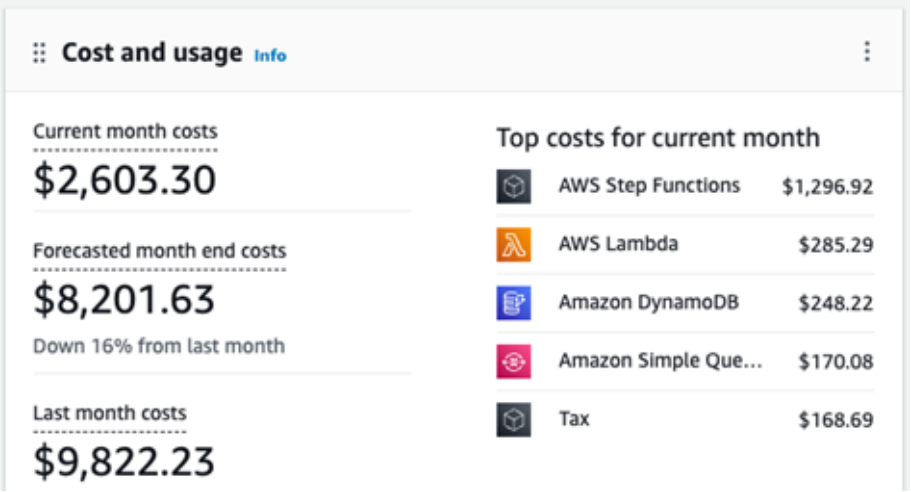
Steep learning curve: education and support

Documentation and tooling improved a lot since 5 years ago

Cost-benefit of serverless...



Search Alerts



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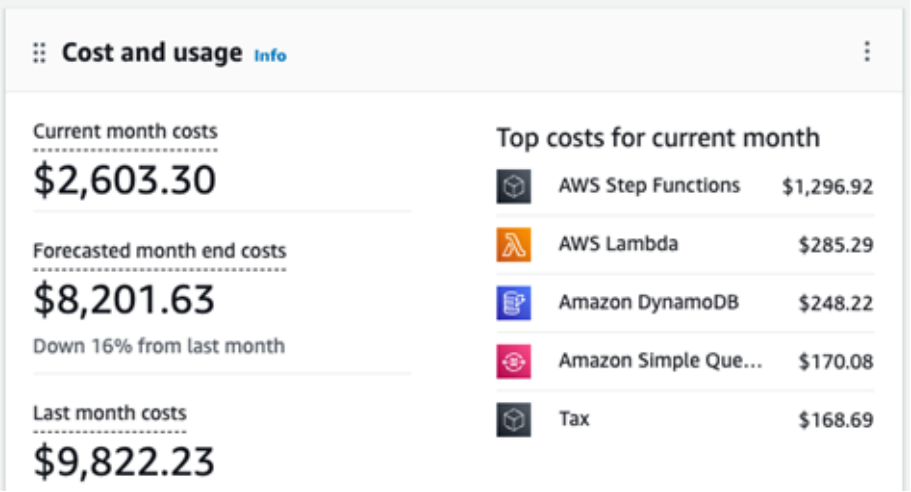


Volume:
~2 million notifications per day

< $0.14 per 1000 notifications

General costs perspective

Fairly predictable and ~~easy~~ possible to calculate

Lambda: ~5% of the total costs

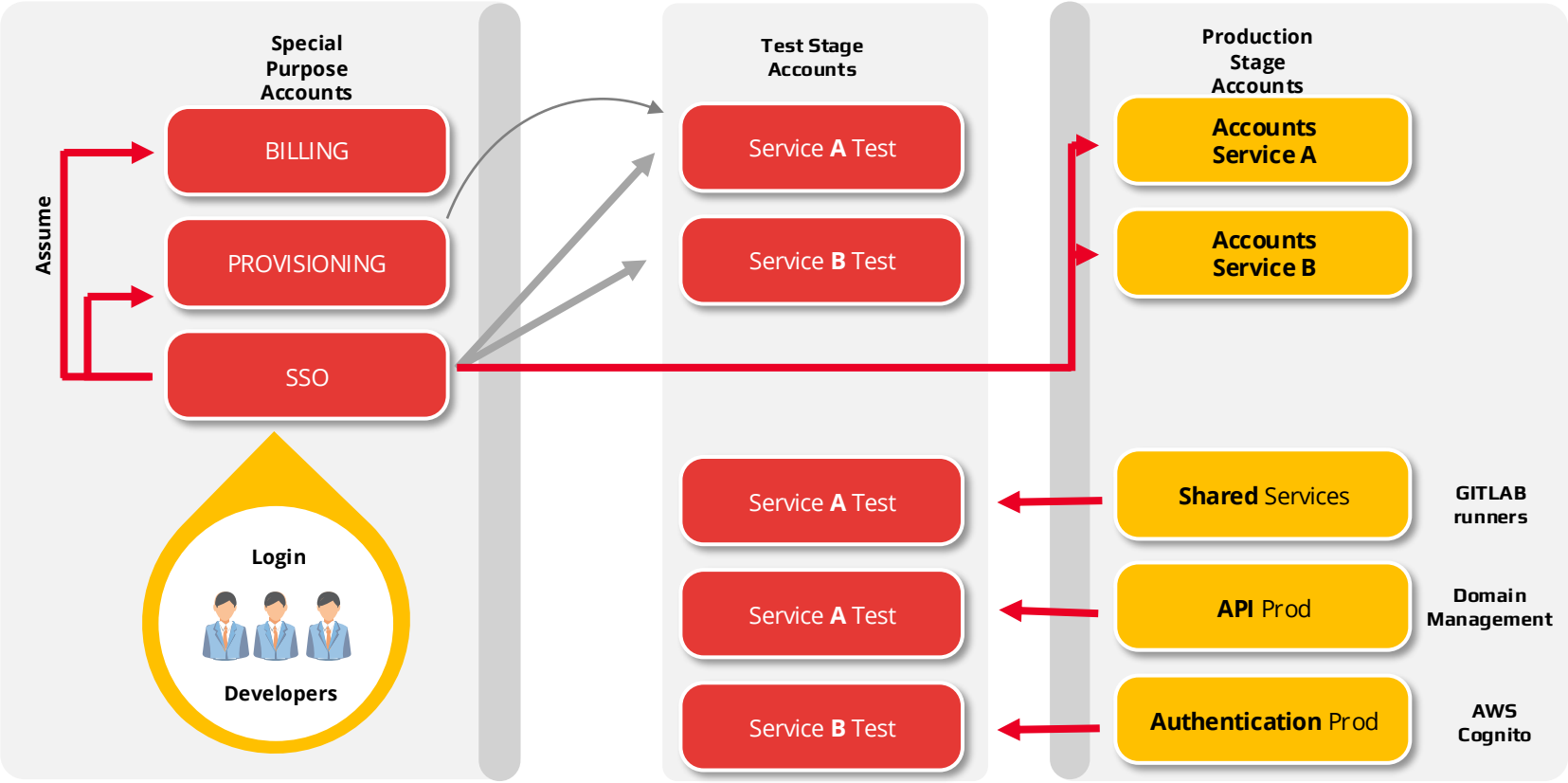
Step Functions: ~8% of the total costs

Other main costs drivers:

- RDS & DynamoDB
- Cloudwatch
- S3

Stable pricing (sometimes get **cheaper**)

Multi-Account Setup



Budget alarms and cost-effective design

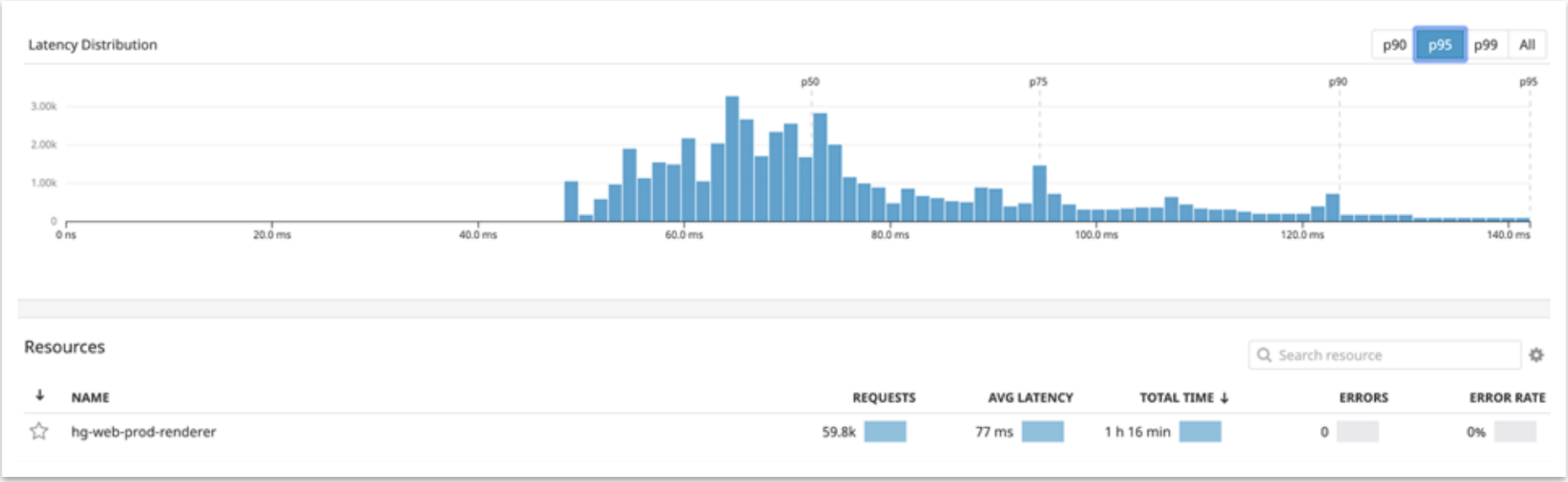
Undeniable advantages

- Security and cost-effectiveness.
  Shared responsibility model and pay-as-you-go model

- Real-time scalability and reliability.
  Happiest on-call team in town

- Nearly infinite development boxes.
  No surprises on production

Server-side rendered listing page



Business impact

- Availability and traffic management.
  State-of-the-art bot control and latency improvements (w/ SEO benefits)

- Shift in observability focus.
  More business-centric, less infrastructure-centric

- User-centric design mindset with faster iterations.
  Worry less about tech, more about impact

Business impact

- Faster and scalable processes.
  Listings take seconds to be available online and notifications sent

- Time to market.
  Many new features released while reengineering (including hackathons)

- Developer experience.
  Time dedicated to infrastructure is shifted to relevant improvements

Serverless is not a hammer

- We're not snowflakes: many are going through similar challenges.
- When you start tinkering too much, look for alternatives.
- Infrastructure as Code is crucial: choose wisely.
- Carefully handle back pressure in event-driven services.

Questions?